

- RAG Learning

4.1 Angles, Polygons & Constructions

Pearson Edexcel GCSE Mathematics 1MA1 — Geometry and measures — spec refs G1–G4, G15 — Papers 1 (non-calculator), 2 and 3 (calculator)

Everything on this page is on **both** the Foundation and Higher tiers of 1MA1. Higher papers simply use these facts in longer, multi-step problems with less guidance.

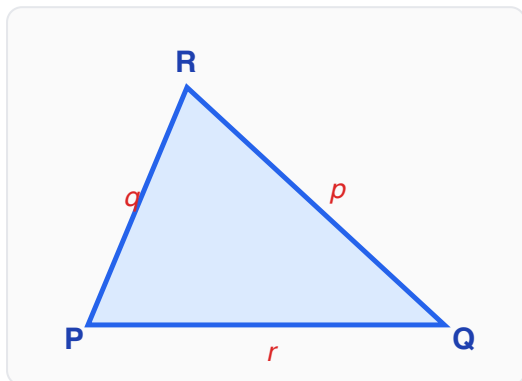
4.1.1 Geometric Terms and Notation (G1)

Term	Meaning
Point, line, plane	A point marks a position; a line is straight (a line segment has two end points); a plane is a flat surface.
Vertex (plural vertices)	A corner where sides or edges meet.
Edge	A line where two faces of a 3D solid meet.
Parallel lines	Lines in the same direction that never meet; shown with matching arrow marks.
Perpendicular lines	Lines that cross or meet at 90° ; the right angle is shown with a small square.
Polygon	A closed 2D shape with straight sides.
Regular polygon	A polygon with all sides equal and all angles equal.
Line (reflection) symmetry	A mirror line divides the shape into two halves that are reflections of each other.

Rotational symmetry

The order is how many times the shape fits onto itself in one full turn. A regular n -gon has n lines of symmetry and rotational symmetry of order n .

Labelling triangles and angles



- Vertices take capital letters: triangle PQR.
- The side opposite angle P is written p , opposite Q is q , opposite R is r .
- Angle PQR (also $\angle PQR$ or $P\hat{Q}R$) is the angle **at Q**, between QP and QR. The middle letter is always the vertex.
- The line segment joining P and Q is written PQ.

Exam Tip: Edexcel questions sometimes give a description with no diagram ("ABCD is a kite with $AB = AD$..."). Sketch it straight away, label the vertices in order round the shape, and mark equal sides, parallel sides and right angles before you calculate anything.

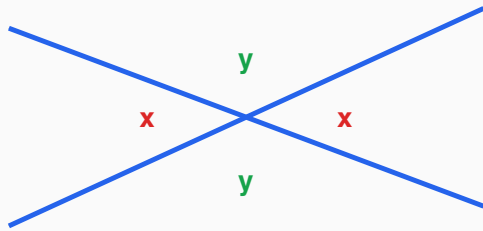
4.1.2 Basic Angle Facts (G3)

Type of angle	Size
Acute	Less than 90°
Right angle	Exactly 90°
Obtuse	More than 90° but less than 180°
Reflex	More than 180° but less than 360°

Angles on a straight line add up to 180°

Angles at a point add up to 360°

Vertically opposite angles are equal



$x + y = 180^\circ$; vertically opposite angles equal

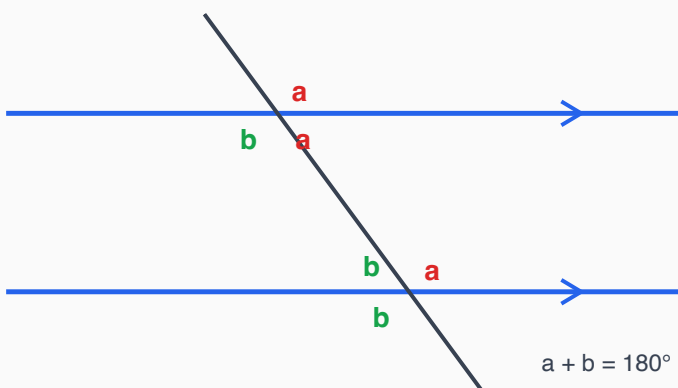
Worked Example: Two angles on a straight line are $4x$ and $5x$. Work out the size of the larger angle.

$$4x + 5x = 180 \Rightarrow 9x = 180 \Rightarrow x = 20$$

$$\text{Larger angle} = 5 \times 20 = \mathbf{100^\circ}$$

4.1.3 Angles and Parallel Lines (G3)

A line crossing a pair of parallel lines is a **transversal**. It makes the same set of angles at each parallel line, which gives three key relationships.



$$a + b = 180^\circ$$

Alternate angles are equal — between the parallel lines, on opposite sides of the transversal.

Corresponding angles are equal — in the same position at each parallel line.

Co-interior angles add up to 180° — between the parallel lines, on the same side of the transversal.

Exam Tip: When a question says "give reasons", Pearson mark schemes require the angle fact written in full, such as "alternate angles are equal". Shorthand like "Z angles" or "F angles" is not accepted.

Worked Example: A transversal crosses two parallel lines. Two corresponding angles are $(3x + 5)^\circ$ and $(5x - 37)^\circ$. Work out x .

Corresponding angles are equal: $3x + 5 = 5x - 37$
 $42 = 2x \Rightarrow x = 21$ (each angle is 68°)

Worked Example: PQ and RS are parallel with Q and S on the right. Point T lies between them, to the right of P and R. Angle QPT = 48° and angle SRT = 27° . Work out angle PTR.

Draw a line through T parallel to PQ and RS.

Angle between TP and this line = 48° (alternate angles are equal).

Angle between TR and this line = 27° (alternate angles are equal).

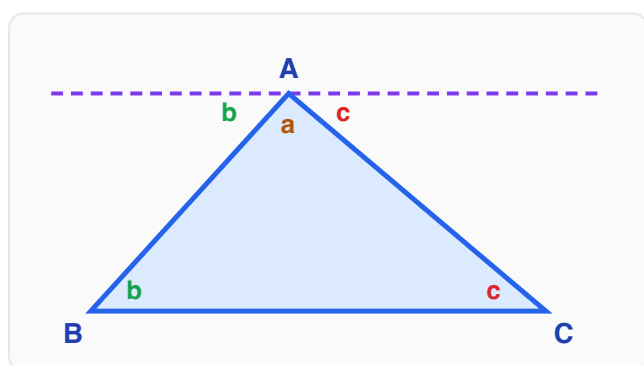
Angle PTR = $48 + 27 = 75^\circ$

4.1.4 Triangles (G3, G4)

Triangle	Properties
Equilateral	3 equal sides, 3 angles of 60° , 3 lines of symmetry, rotational symmetry of order 3

Isosceles	2 equal sides, the 2 base angles opposite them are equal, 1 line of symmetry
Scalene	All sides different, all angles different
Right-angled	One angle of 90°
Acute-angled	Every angle less than 90°
Obtuse-angled	One angle more than 90°

Deriving the angle sum of a triangle



Worked Example: Through A draw the line parallel to BC.

On the left of angle a the angle is b (alternate angles are equal, using transversal AB).

On the right of angle a the angle is c (alternate angles are equal, using transversal AC).

The three angles b , a , c make a straight line, so $a + b + c = 180^\circ$. ■

Exterior angle of a triangle

If one side of a triangle is extended, the **exterior angle** equals the **sum of the two interior opposite angles**.

Reason: $a + b + c = 180$ (angles in a triangle) and $c + e = 180$ (angles on a straight line), so $e = a + b$.

Worked Example: The angles of a triangle are x , $2x + 10$ and $3x - 40$ (in degrees). Work out each angle.

$$x + 2x + 10 + 3x - 40 = 180 \Rightarrow 6x - 30 = 180 \Rightarrow x = 35$$

Angles: **35°** , **80°** and **65°** (check: $35 + 80 + 65 = 180$ ✓)

4.1.5 Special Quadrilaterals (G4)

Any quadrilateral can be split by a diagonal into two triangles, so its angles add up to $2 \times 180^\circ = 360^\circ$.

Shape	Sides	Angles	Diagonals	Symmetry (lines / rotation)
Square	4 equal; opposite sides parallel	All 90°	Equal; bisect each other at 90°	4 / order 4
Rectangle	Opposite sides equal and parallel	All 90°	Equal; bisect each other	2 / order 2
Rhombus	4 equal; opposite sides parallel	Opposite angles equal	Bisect each other at 90°	2 / order 2
Parallelogram	Opposite sides equal and parallel	Opposite angles equal; neighbouring angles add to 180°	Bisect each other	0 / order 2
Kite	Two pairs of equal adjacent sides	One pair of opposite angles equal	Meet at 90° ; one bisects the other	1 / order 1

Trapezium	Exactly one pair of parallel sides	Co-interior angles between the parallel sides add to 180°	No special property (equal in an isosceles trapezium)	0 / order 1 (isosceles trapezium: 1 line)
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Worked Example: The angles of a quadrilateral are $3x$, $2x + 15$, $x + 45$ and $4x$ (in degrees). Work out the largest angle.

$$10x + 60 = 360 \Rightarrow 10x = 300 \Rightarrow x = 30$$

Angles: 90° , 75° , 75° , 120° . Largest angle = **120°**

Worked Example: ABCD is a parallelogram with angle $ABC = 68^\circ$. Work out angle BCD and angle CDA.

AB is parallel to DC, so angles ABC and BCD are co-interior: angle BCD = $180 - 68 =$
 112°

Opposite angles of a parallelogram are equal: angle CDA = angle ABC = **68°**

Exam Tip: Shapes sit inside families: every square is a rectangle and a rhombus, and every rhombus is a parallelogram. If asked to give a property that distinguishes two shapes, pick one the other shape genuinely lacks — e.g. a rectangle's diagonals are equal, a general parallelogram's are not.

4.1.6 Angles in Polygons (G3)

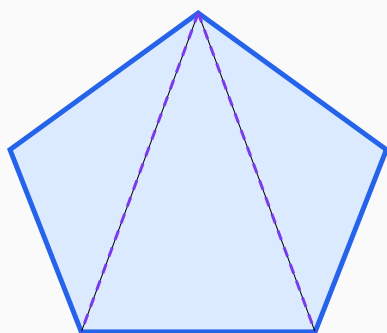
Sides	Name	Interior angle sum	Each interior angle (regular)	Each exterior angle (regular)
3	Triangle	180°	60°	120°
4	Quadrilateral	360°	90°	90°

5	Pentagon	540°	108°	72°
6	Hexagon	720°	120°	60°
8	Octagon	1080°	135°	45°
9	Nonagon	1260°	140°	40°
10	Decagon	1440°	144°	36°

Sum of interior angles of an n -sided polygon $= (n - 2) \times 180^\circ$

Sum of exterior angles of **any** polygon $= 360^\circ$

Regular polygon: exterior angle $= \frac{360^\circ}{n}$, interior angle $= 180^\circ - \text{exterior angle}$



From one vertex: $5 - 2 = 3$ triangles, $3 \times 180^\circ = 540^\circ$

Worked Example: Each interior angle of a regular polygon is 170° . Work out the number of sides.

$$\text{Exterior angle} = 180 - 170 = 10^\circ$$

$$n = 360 \div 10 = \mathbf{36} \text{ sides}$$

Worked Example: The interior angles of a polygon add up to 2520° . How many sides does it have?

$$(n - 2) \times 180 = 2520 \Rightarrow n - 2 = 14 \Rightarrow n = \mathbf{16}$$

Worked Example: Show that a regular polygon cannot have interior angles of 130° .

Exterior angle = $180 - 130 = 50^\circ$, so $n = 360 \div 50 = 7.2$.

The number of sides must be a whole number, so no regular polygon has interior angles of 130° .

Exam Tip: Go via the exterior angle whenever you can — on Paper 1, $360 \div 10$ is far quicker and safer than solving $(n - 2) \times 180 = 170n$.

4.1.7 Multi-step Angle Reasoning (G3, G4)

Longer problems combine several facts: polygon angles, isosceles triangles and parallel lines. Work one angle at a time, write each angle on the diagram, and give the reason beside each step.

Worked Example: ABCDEF is a regular hexagon. Work out angle ACD.

Interior angle of a regular hexagon = $(6 - 2) \times 180 \div 6 = 120^\circ$.

$AB = BC$ (sides of a regular hexagon), so triangle ABC is isosceles and angle $BCA = (180 - 120) \div 2 = 30^\circ$ (base angles equal, angles in a triangle add to 180°).

Angle $ACD = \text{angle } BCD - \text{angle } BCA = 120 - 30 = \mathbf{90^\circ}$

Worked Example: Triangle XYZ has $XY = XZ$. The line through X parallel to YZ makes an angle of 54° with XY. Work out angle YXZ.

Angle $XYZ = 54^\circ$ (alternate angles are equal).

Angle $XZY = 54^\circ$ (base angles of an isosceles triangle are equal).

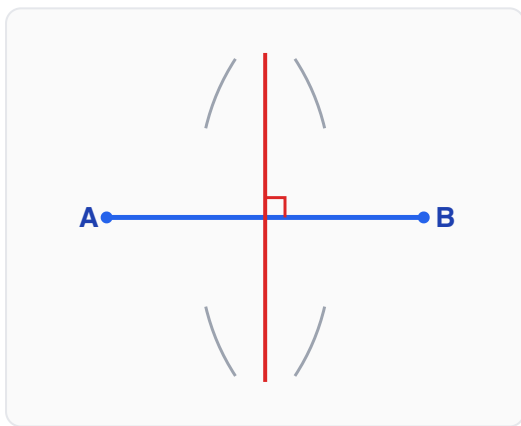
Angle $YXZ = 180 - 54 - 54 = \mathbf{72^\circ}$ (angles in a triangle add up to 180°)

Exam Tip: "Give reasons" questions on Edexcel papers typically award a separate mark for a full set of correct reasons. Name the triangle that is isosceles and say why (e.g. "AB = BC, sides of a regular hexagon").

4.1.8 Ruler and Compass Constructions (G2)

You need a sharp pencil, a ruler and a pair of compasses. Do not use a protractor unless the question allows it. **Every construction arc must be left visible** — the arcs are the evidence the examiner is looking for.

Perpendicular bisector of a line segment AB



1. Set the compasses to a radius greater than half of AB.
2. Put the point on A and draw an arc above AB and an arc below it.
3. Without changing the radius, do the same from B so that the arcs cross in two places.
4. Draw a straight line through the two crossing points. It cuts AB at its midpoint, at 90° .

Perpendicular from a point P to a line

1. With the compass point on P, draw an arc that crosses the line at two points, X and Y.
2. From X and then from Y (same radius), draw arcs on the opposite side of the line from P so they cross.
3. Join P to the crossing point. This line meets the original line at 90° .

Perpendicular at a point P on a line

1. With centre P, mark two points X and Y on the line, the same distance either side of P.
2. Widen the compasses and draw arcs from X and from Y that cross above P.
3. Join P to the crossing point.

The **shortest distance** from a point to a line is the **perpendicular distance**: the length of the line segment from the point that meets the line at 90° .

Bisecting an angle

1. With centre at the vertex O, draw an arc crossing both arms of the angle, at X and Y.
2. From X and from Y, using the same radius, draw two arcs that cross at Z inside the angle.
3. Draw the line OZ. It divides the angle into two equal angles.

Constructing angles of 60° , 30° , 90° and 45°

1. For 60° : mark O on a line, draw an arc centred on O cutting the line at X, then with the **same radius** draw an arc centred on X cutting the first arc at Y. Triangle OXY is equilateral, so angle XOY = 60° .
2. For 30° : bisect the 60° angle.
3. For 90° : construct the perpendicular at a point on a line. For 45° : bisect the 90° angle.

Exam Tip: Rubbing out construction arcs normally costs the marks even if the final line is correct. Edexcel usually allows a tolerance of about 2 mm for lengths and 2° for angles.

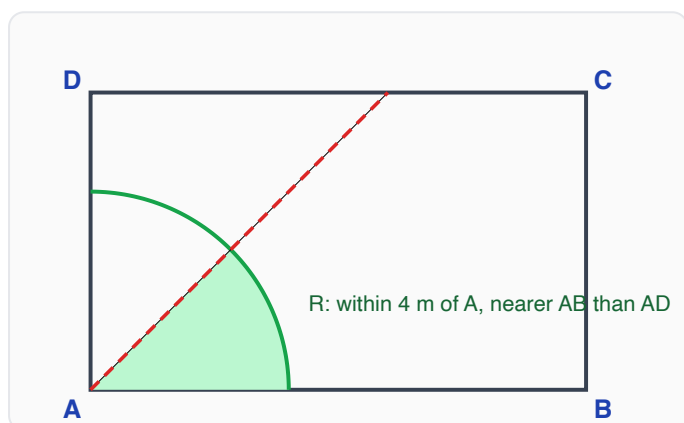
4.1.9 Loci (G2)

A **locus** (plural loci) is the set of all points that satisfy a rule.

Rule	Locus
A fixed distance from a point P	A circle with centre P
A fixed distance from a line segment AB	Two lines parallel to AB, joined by a semicircle at each end

Equidistant from two points A and B	The perpendicular bisector of AB
Equidistant from two lines that meet	The bisector of the angle between them

Regions use inequality language: "less than 3 m from P" is the inside of the circle; "nearer to A than to B" is the side of the perpendicular bisector containing A; "nearer to AB than to AD" is the side of the angle bisector containing AB. When there are several conditions, shade only the region where they all hold, and label it R.



Worked Example: A rectangular yard ABCD is drawn to a scale of 1 cm to 2 m. A lamp must be within 8 m of A and nearer to AB than to AD. How do you show the region?

8 m at 1 cm to 2 m is $8 \div 2 = 4$ cm: draw an arc of radius 4 cm, centre A.

Construct the bisector of angle DAB with compasses.

Shade the region inside the arc and on the AB side of the bisector (as in the diagram).

4.1.10 Measuring, Scale Drawings and Bearings (G15)

Measuring and scale drawings

- Measure lengths to the nearest millimetre. Measure angles to the nearest degree, reading the protractor scale that starts at 0° on the arm you are measuring from.

- A map scale can be written as 1 cm to 500 m or as a ratio such as 1 : 50 000 (1 cm on the map is 50 000 cm in real life).
- Real length = map length $\times$ scale factor; map length = real length $\div$ scale factor. Then convert:
100 000 cm = 1 km.

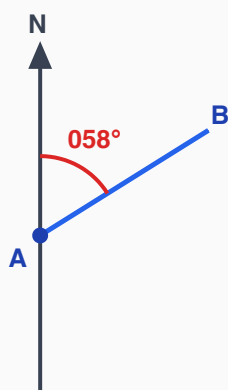
Worked Example: A map has scale 1 : 40 000. Two stations are 7.5 cm apart on the map. Work out the real distance in kilometres.

$$7.5 \times 40\,000 = 300\,000 \text{ cm} = 3000 \text{ m} = \mathbf{3 \text{ km}}$$

Worked Example: On the same map, how far apart are two farms that are really 1.8 km apart?

$$1.8 \text{ km} = 180\,000 \text{ cm, and } 180\,000 \div 40\,000 = \mathbf{4.5 \text{ cm}}$$

Bearings



A bearing is:

- measured from **North**
- measured **clockwise**
- written with **three figures** (e.g. 058°, 305°)

Back bearing: add 180° if the bearing is less than 180°, otherwise subtract 180°

Compass point	N	NE	E	SE	S	SW	W	NW
Bearing	000°	045°	090°	135°	180°	225°	270°	315°

Worked Example: The bearing of B from A is 058°. Work out the bearing of A from B.

$058 + 180 = \mathbf{238}^\circ$ (the North lines at A and B are parallel, so co-interior angles add to 180°)

Worked Example: A boat sails from P on a bearing of 040° to A. It then sails on a bearing of 130° to B. Show that angle PAB = 90° .

Bearing of P from A = $040 + 180 = 220^\circ$

Angle PAB = $220 - 130 = \mathbf{90}^\circ$

Exam Tip: Read "the bearing of X **from** Y" as: stand at Y, face North, turn clockwise until you face X. Draw the North line at Y before measuring, and always write three figures — 58° should be written 058° .